Intern Research Engineer - AI Agent Systems

Job description

Huawei Canada has an immediate internship opening for a Research Engineer.

About the Team:

The Technology Planning and Cooperation Department promotes strategic innovation across all of Huawei Canada. Working with fellow experts, the focus is researching new and upcoming areas of technology as well as strategic planning to help achieve long term vision of Huawei. This department offers a unique opportunity to leverage both technical and business skills.

About the job:
• Architect and implement high-scalability AI Agent pipelines, transforming complex requirements into reliable execution structures (e.g., TaskGraph IR, Operator DAGs, and multi-agent coordination frameworks).
• Develop and optimize automated code generation, reasoning, and self-debugging loops that interact directly with multi-framework environments (e.g., creating bridge layers, compilation pipelines, or automated code refactoring tools).
• Design and optimize advanced memory substrates for Agents, tackling challenges related to long-context handling, state persistence, and execution efficiency.
• Establish rigorous, automated evaluation frameworks to benchmark Agent execution accuracy, latency, and reliability under non-deterministic LLM behaviors.
• Translate the latest research in Agentic workflows (e.g., ReAct, Reflection, Planning, Tool Use) into production-grade, highly performant software libraries.

The total target annual compensation (based on 2,080 hours per year) ranges from $58,000 to $104,000 depending on education, experience, and demonstrated expertise.

About the ideal candidate:
• Currently pursuing a Master’s or Bachelor’s Degree in Computer Science or a highly quantitative field, with a focus on AI, Software Engineering Systems, or Distributed Systems.
• Proficient in orchestrating LLMs programmatically. Deep understanding of Agentic design patterns (Planning, Memory, Tool Use, Multi-Agent Collaboration).

• Exceptional coding skills in Python and/or TypeScript/Node.js. Familiarity with structural design patterns and engineering clean, maintainable codebases.
• Deep experience with frameworks such as LangChain, LlamaIndex, LangGraph, CrewAI, AutoGen, or building custom agentic runtime systems.
• Ability to quickly digest, implement, and improve upon cutting-edge ArXiv papers on LLM reasoning and code generation.
• Contributions to open-source AI Agent frameworks, developer tools, or compiler-related projects is an asset.
• Experience with Code-LLM optimization (e.g., custom fine-tuning for code generation, structured outputs), embedded database engines or high-performance vector stores is an asset.
• Familiarity with cross-framework mobile compilation or bridge layers (e.g., Android/AOSP or hybrid application runtimes) is an asset.

Additional Information:

Huawei Canada is committed to a fair, inclusive, and accessible recruitment process. If you require accommodation during any stage of the hiring process, please let us know and we will work with you to meet your needs.

All applications for this position are reviewed directly by our hiring team, we do not use artificial intelligence tools to screen or select candidates.